ADS1298 #1

Electrodes

E1 R1 2KΩ C1 620pF GND; R2 2KΩ C2 620pF GND

E2 R3 2KΩ C3 620pF GND; R4 2KΩ C4 620pF GND

E3 R5 2KΩ C5 620pF GND; R6 2KΩ C10 620pF GND

E4 R7 2KΩ C11 620pF GND; R8 2KΩ C12 620pF GND

E5 R9 2KΩ C13 620pF GND; R10 2KΩ C14 620pF GND

E6 R11 2KΩ C15 620pF GND; R12 2KΩ C16 620pF GND

E7 R13 2KΩ C17 620pF GND; R14 2KΩ C18 620pF GND

E8 R15 2KΩ C21 620pF GND; R16 2KΩ C22 620pF GND

+3.3V AVDD C6 0.1uF AGND; C7 10uF AGND

DVDD +3.3V C8 0.1uF AGND; C9 10uF AGND

U1 ADS1298xPAG

IN1P IN1N IN2P IN2N IN3P IN3N IN4P IN4N IN5P IN5N IN6P IN6N IN7P IN7N IN8P IN8N

AVDD AVDD1 DVDD

WCT RLDOUT RLDIN RLDINV RLDREF

AVDD

TESTP_PACE_OUT1 TESTP_PACE_OUT2

GPIO1 GPIO2 GPIO3 GPIO4 AGND

DAISY_IN SCLK DIN CS DOUT

Unused, shorted AGND

SCLK MOSI CS_ADC1 MISO

CLK CLKSEL START DRDY PWDN RESET

U1 CLK_EN=1 required

ADC_CLK DVDD START DRDY_ADC1 PWDN RESET

RESV1 AGND

VCAP1_ADC1 VCAP2_ADC1 VCAP3_ADC1 VCAP4_ADC1

VCAP1 VCAP2 VCAP3 VCAP4

VREFP_ADC1 C19 0.1uF C20 10uF AGND

VREFP VREFN AVSS AVSS1 AGND DGND GND

C23 22uF AGND; C24 1uF AGND; C25 1uF AGND; C26 1uF AGND

ADS1298 #2

Electrodes

E9 R17 2KΩ C27 620pF GND; R18 2KΩ C28 620pF GND

E10 R19 2KΩ C29 620pF GND; R20 2KΩ C30 620pF GND

E11 R21 2KΩ C31 620pF GND; R22 2KΩ C36 620pF GND

E12 R23 2KΩ C37 620pF GND; R24 2KΩ C38 620pF GND

E13 R25 2KΩ C39 620pF GND; R26 2KΩ C40 620pF GND

E14 R27 2KΩ C41 620pF GND; R28 2KΩ C42 620pF GND

E15 R29 2KΩ C43 620pF GND; R30 2KΩ C44 620pF GND

E16 R31 2KΩ C47 620pF GND; R32 2KΩ C48 620pF GND

+3.3V AVDD C32 0.1uF AGND; C33 10uF AGND

DVDD +3.3V C34 0.1uF AGND; C35 10uF AGND

U2 ADS1298xPAG

IN1P IN1N IN2P IN2N IN3P IN3N IN4P IN4N IN5P IN5N IN6P IN6N IN7P IN7N IN8P IN8N

AVDD AVDD1 DVDD

WCT RLDOUT RLDIN RLDINV RLDREF

AVDD

TESTP_PACE_OUT1 TESTP_PACE_OUT2

GPIO1 GPIO2 GPIO3 GPIO4 AGND

DAISY_IN SCLK DIN CS DOUT

Unused, shorted AGND

SCLK MOSI CS_ADC2 MISO

CLK CLKSEL START DRDY PWDN RESET

U1 CLK_EN=1 required

ADC_CLK START DRDY_ADC2 PWDN RESET

F09: DRDY_ADC2 only; MCU still uses U1 DRDY

RESV1 AGND

VCAP1_ADC2 VCAP2_ADC2 VCAP3_ADC2 VCAP4_ADC2

VCAP1 VCAP2 VCAP3 VCAP4

VREFP_ADC2 C45 0.1uF C46 10uF AGND

VREFP VREFN AVSS AVSS1 AGND DGND GND

C49 22uF AGND; C50 1uF AGND; C51 1uF AGND; C52 1uF AGND

+5V +3.3V DVDD

U3 ESP32-S3-DevKitC

CHIP_PU 5V 3V3 GND

GPIO43/U0TXD GPIO44/U0RXD

GPIO15/ADC2_CH4/32K_P GPIO16/ADC2_CH5/32K_N GPIO20/USB_D+ GPIO19/USB_D-

GPIO39/MTCK GPIO40/MTDO GPIO41/MTDI GPIO42/MTMS

GPIO0 GPIO1/ADC1_CH0 GPIO2/ADC1_CH1 GPIO3/ADC1_CH2 GPIO4/ADC1_CH3 GPIO5/ADC1_CH4 GPIO6/ADC1_CH5 GPIO7/ADC1_CH6 GPIO8/ADC1_CH7 GPIO9/ADC1_CH8 GPIO10/ADC1_CH9 GPIO11/ADC2_CH0 GPIO12/ADC2_CH1 GPIO13/ADC2_CH2 GPIO14/ADC2_CH3 GPIO17/ADC2_CH6 GPIO18/ADC2_CH7 GPIO21 GPIO35 GPIO36 GPIO37 GPIO38 GPIO45 GPIO46 GPIO47 GPIO48

RESET PWDN START DRDY_ADC1 CS_ADC1 MOSI SCLK MISO CS_ADC2

ESP32-S3-DevKitC

AGND

F21: probe access only; test-pad footprints/layout pending

TP1 ADC_CLK — ADC_CLK
TP2 DRDY_ADC1 — DRDY_ADC1
TP3 DRDY_ADC2 — DRDY_ADC2
TP4 3V3 — +3.3V
TP5 BAT — +3.7V
TP6 AGND — AGND
TP7 VREFP_ADC1 — VREFP_ADC1
TP8 VCAP1_ADC1 — VCAP1_ADC1
TP9 VCAP2_ADC1 — VCAP2_ADC1
TP10 VCAP3_ADC1 — VCAP3_ADC1
TP11 VCAP4_ADC1 — VCAP4_ADC1
TP12 VREFP_ADC2 — VREFP_ADC2
TP13 VCAP1_ADC2 — VCAP1_ADC2
TP14 VCAP2_ADC2 — VCAP2_ADC2
TP15 VCAP3_ADC2 — VCAP3_ADC2
TP16 VCAP4_ADC2 — VCAP4_ADC2

Grounding Configuration

AGND GND

ADC LDO Configuration

U4 LP2985-3.3

+3.7V C53 1uF GND; EN; VIN ON/OFF GND BP VOUT; +3.3V C54 2.2uF GND; C55 10nF GND

Battery Configuration

+3.7V BT1 Battery GND

MCU LDO Configuration

U5 TLV75533PDBV

+3.7V C56 1uF GND; EN; IN EN GND OUT; +3.3V C57 200uF GND

START R33 10k AGND

RESET R34 10k AGND

PWDN R35 10k AGND

F09: 10k pulldowns bias START, RESET and PWDN low while MCU is high-Z.
CS defaults / complete power sequencing remain unresolved.

V2 PARTIAL CORRECTIONS — NOT READY FOR POWER-UP OR WEARER USE
F04/F05: parallel LDO outputs and floating EN remain; power architecture requires approval.
F06/F18: restore U1 CLK_EN after RESET; allow U2 clock/POR; configure both before shared START.
F09: START/RESET/PWDN biased low while MCU is high-Z; CS defaults and full sequencing remain unresolved.
Enable/settle 2.4V reference; verify gain/rate; keep PACE/RLD/WCT and RLD rerouting off.
F07/F08: protected body bias and wearer protection remain unimplemented.

Version 1.2
McGill AI Lab — Researcher: Karen Chen Lai
Sheet: /
File: open-semg-16-v2.kicad_sch
Title: sEMG Wirstband
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.7 | Id: 1/1